7

Acquisition of (functional) morphology

Functional morphology is considered to be the locus of language variation.[1] It hosts the features that regulate how a specific language grammar functions. Functional morphology has three essential aspects: a) it has morphophonological form (it is pronounced and is frequently attached to another word as a prefix, suffix, or infix); b) it carries grammatical meaning (e.g., tense, aspect, gender, case); and c) it has syntactic consequences by regulating which phrases move around in the sentence and which stay where they merged. After learning the vocabulary items of the second language, the functional morphology is the single most important form–function mapping that learners have to master. Without the inflectional affixes and the function words, the message of a sentence may be comprehensible in communication, but the sentence is not well formed and no native speaker would produce it that way.

[1] See Chapters 1 and 2 for the language architecture and how that affects acquisition.

Since functional morphology is visible, some of it even salient, and highly frequent, it can be tested in naturalistic or elicited production. The most common method of evaluating knowledge of morphology has been to use a sample of transcribed learner speech in order to count the environments in which a native speaker of the language would use a particular piece of morphology. These are known as obligatory contexts. For example, a plural context is a noun phrase of this sort: *two book___*. A speaker of English would produce *two book-s*, while a learner of English might drop the *-s*. An obligatory context for the progressive *-ing* suffix would be *I am go___ to the market*. The researcher counts how many times the individual learner produced the appropriate morphology in the obligatory context. The result should be reported both as a raw count and a percentage.

Why might it be important to report accuracy in obligatory contexts both in raw count and as a percentage?

However, one should never disregard meaning. The features that regulate meanings are called "interpretable features," and we will see them exemplified in this chapter. Meaning is not visible on the surface, and we can't directly record meaning. Of necessity, we need to create experimental situations to extract speakers' interpretations. One clue as to what meaning learners attribute to a morpheme may be appropriate usage in production, but this clue can also be misleading. Extensive research has indicated that children and other learners might not attribute an adultlike interpretation to function words and morphemes, even if they produce them correctly. For example, it is well known that while English-speaking children produce correct object pronouns as in *Mama Bear is touching* ***her***, half of the time they actually interpret such a sentence to mean *Mama Bear is touching* ***herself*** (Chien and Wexler 1990), which is not at all the same meaning, is it? Therefore, caution should be used in assuming that if a morpheme is used correctly, it is actually fully acquired.

7.1 Morpheme studies

The acquisition of functional morphology has been of great interest to L2 researchers since the 70s. This interest was sparked by Roger Brown's seminal work on child language acquisition (Brown 1973), which we reviewed in Chapter 5. In this work, Brown tracks the grammatical developments in three children's speech by counting the correct usage of what he called "functor" morphemes, or grammatical morphemes. He established that Adam, Eve, and Sarah acquired these morphemes in the same order, though not at the same age. Until that time, work on native language transfer in SLA had taken a behaviorist perspective[2] and engaged in contrastive error analysis. Inspired by Brown's work from a more psychological, generative perspective, a series of studies in the early 1970s tested the idea that child L2 acquisition follows the same milestones and the same route as child L1 acquisition. Dulay and Burt (1973, 1974) proposed the Creative Construction Hypothesis, according to which children acquiring a second language were guided by the innate principles of grammar, causing them to formulate "certain types of hypotheses about the language system being acquired, until the mismatch between what they are exposed to and what they produce is resolved" (1974: 37). This hypothesis rings true even today.

The data supporting this hypothesis came from a standardized test of English (later extended to Spanish), known as the Bilingual Syntax Measure, whose purpose was to elicit functional morphology and evaluate the accuracy of its use in required contexts. The method used seven cartoonlike pictures and 33 questions. For example, the experimenter pointed to two houses in one of the pictures and asked "What are these?" The expected answer would include a plural morpheme as in *house-s*. In one experimental study, Dulay and Burt (1973) compared BSM scores from 5 to 8-year-old Spanish-native children acquiring English under different circumstances and amounts of exposure.[3]

[2] The Behaviorist (behavioral) approach asserts that languages are learned in the same way as any other behavior. Language acquisition is habit formation, so existing habits could be reinforced through imitation, conditioning, stimulus, and response (Skinner 1957). As discussed above, contrastive error analysis was a method of explaining why some features of a second language were more difficult than others to acquire. This method was prevalent in the 1960s and early 1970s, couched in the behaviorist view of language acquisition popular at the time. Difficulty depended on the distance between the learners' native language and the language they were trying to learn.

[3] The children in the Sacramento group had the most exposure to English, as they were born in the US. The San Ysidro group of children were exposed to English only in school, as they went back to their homes in Tijuana every day after school. The Puerto Rican children in East Harlem were recent immigrants and had not been in the US for more than a year. They were exposed to a dual immersion program in school, where subject matter was taught both in Spanish and in English.

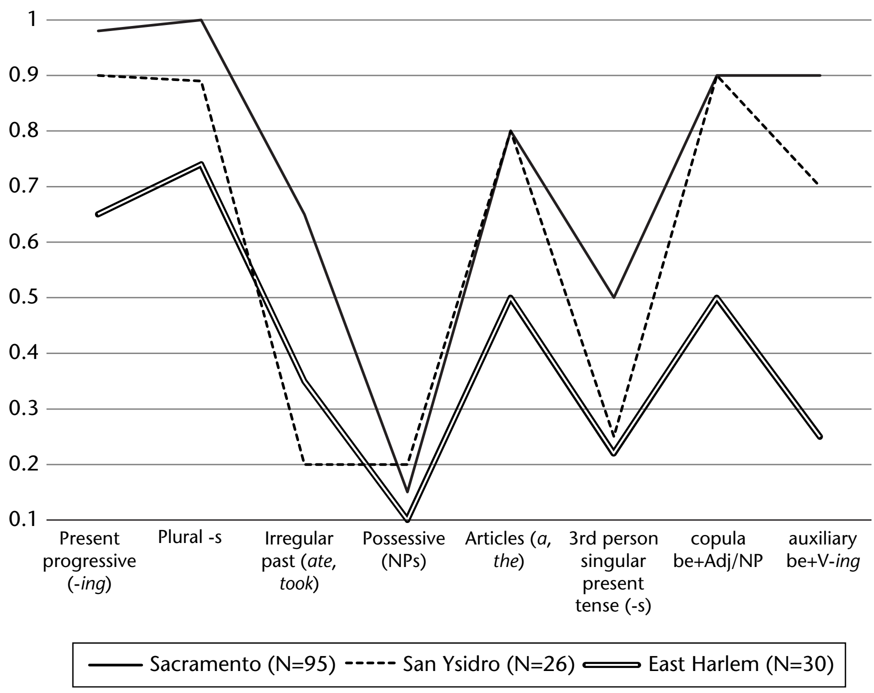


Figure 7.1 Comparison of BSM scores on English L2 functors by three groups of Spanish-speaking children, from Dulay and Burt (1973)

In Figure 7.1, the functor morphemes are plotted on the X axis (not all of Brown's functors were tested), while the mean functor ratios are plotted on the Y axis. The latter were calculated as follows: for each functor, the denominator was the total number of obligatory occasions for that functor for all children across a group, and the nominator was the sum of the scores obtained for each obligatory occasion. What is immediately visible is that although the composite ratios differ among the groups, they vary largely in consort. Look, for instance, at the low ratios on the possessive morpheme.

Even more astonishing, and with a slightly higher number of functors, are the findings of Dulay and Burt (1974). Figure 7.2 illustrates the same congruence of accuracy on the functors for two groups of children from very different linguistic backgrounds: Chinese and Spanish, acquiring English. In this study, Dulay and Burt used two methods of calculating the BSM (group score and group means), only one of which is plotted here.

Bailey, Madden, and Krashen (1974) investigated whether a similar order of acquisition would emerge in adult L2 learners. The learners they tested on

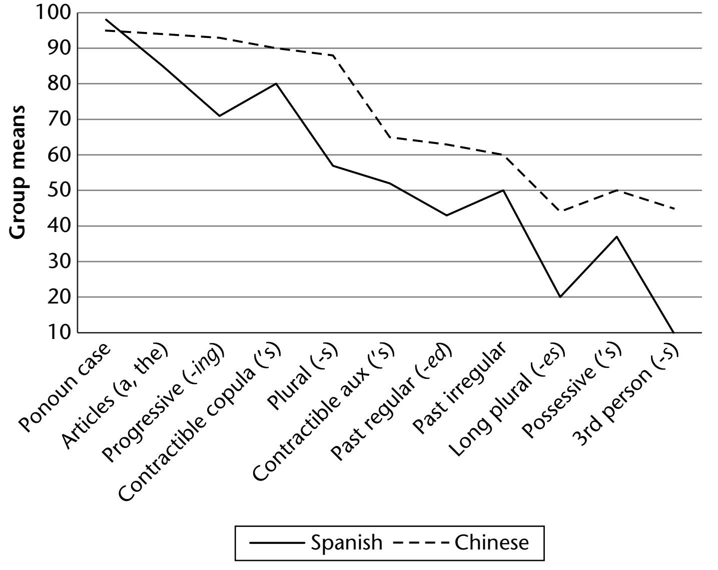


Figure 7.2 Comparison of group means on English L2 functor acquisition by Spanish-speaking and Chinese-speaking children, from Dulay and Burt (1974)

the BSM were divided into Spanish-native and non-Spanish-native, the latter group comprising a variety of typologically different native languages. Such a design was rightly criticized for making it impossible to test the effect of the native language. Bailey et al. discovered that the progressive -*ing*, the contractible copula (*I'm late*), and plural -*s* emerged as the early-acquired morphemes, while possessive '*s* and 3rd person singular subject -*s* were produced with much less accuracy. The order of functor acquisition so established, with slight variation between child L1, child L2, and adult L2 learners, indicated that there could be a certain fixed order to the acquisition of the various functional morphemes (free morphemes and inflectional affixes). The researchers suggested that the fixed order reveals the "creative construction" of a common L2 grammar.

These studies were very influential, although various facets of their methodology were widely criticized. For example, many researchers felt that the morpheme order studies underrepresented the influence of the native language. Luk and Shirai (2009), for instance, uncovered that the order of English morphemes was different for native speakers of Spanish, on the one hand, and for speakers of Japanese, Chinese, and Korean, on the other. The original studies also sparked a lot of replications and new research trying to elucidate why the order of acquisition is as it is (see Exercise 7.1 for various explanations of the order proposed in the vast literature on the issue).

The most substantive criticism of the early morpheme studies concerns the interpretation of the findings. Do accuracy orders really reflect developmental sequences? Some doubt on this strong claim is cast by the realization that if a learner accurately supplies a specific morpheme in all the required contexts, but also overuses the same morpheme in contexts where native speakers would not use it, this learner would score 100% on the BSM, while her successful acquisition of the morpheme would be suspect. One such case was uncovered by Wagner-Gough and Hatch (1975). Five-year-old Homer, an Iranian child learning English, used the *-ing* progressive morpheme early and pervasively, as other children do. However, further inspection of Homer's production shows that he did not contrast this tense–aspect morpheme with other tense–aspect marking forms. He also overused the *-ing* to signal immediate intention, distant future, past events, and imperatives. For the latter functional meaning, see the example below (Wagner-Gough and Hatch 1975: 303).

(1) Imperative meaning:
Okay, sit down over here! *Sitting* down like that!
(= Sit down over here. Sit down like that!)

It is not difficult to improve the BSM scoring method by taking into account extensions of meaning leading to overuse of some morpheme. One such improvement was offered by Stauble (1984). In investigating the verbal morpheme accuracy of six Spanish and six Japanese learners of English in a quasi-longitudinal study, Stauble (1984: 329) used a different method for calculating targetlike usage. First, she counted the accurate suppliance of a morpheme in obligatory context. Say, a learner provided the *-ing* morpheme in five of the ten obligatory contexts, for an accuracy score of 5/10 = 50%. However, the learner also overused the *-ing* on three further occasions, such as in (1) above. Stauble would add 3 to the denominator of the previous calculation, 5/(10+3) = 5/13 = 38%, a more realistic numeric equivalent of the learners' performance on this morpheme.

The discussion of calculating a certain percentage that is purported to reflect a learner's accuracy, that is, targetlike performance on a certain functional morpheme opens a much more significant and fundamental issue. It is related to the question we pondered above: Do accuracy orders really reflect developmental sequences? One might paraphrase it like this:

When can we assume that a certain morpheme has been acquired? Of course, the answer to this question depends on one's assumptions of what "to be acquired" means, for a specific morpheme. Let us continue with the example of *be* + *-ing*, the English progressive tense. Leaving aside modal meanings and vastly oversimplifying, its semantics reflects an ongoing event in the past, present, or future. The function of this aspectual tense requires that it be used when the intended meaning is of a process with its completion left open (*Mary was eating a sandwich when I came in*). With respect to form, both a finite, correctly inflected instance of *be* and an *-ing* participle are required. Correct form–meaning mapping would suggest that the learner *contrasts* this aspectual tense with, for example, the past simple tense (*Mary ate a sandwich*). For such a contrast to be demonstrated, the same learner must use both simple and progressive tenses and must know the meaning difference between them. Only then could one argue that the functional category of Aspect has been activated in the learner's grammar. This formulation is just shorthand for "the learner has acquired the progressive form–meaning mapping, that is, what grammatical form is paired with the ongoing event meaning."

But what percentage of targetlike usage in naturalistic or elicited production should we take to be indicative of successful acquisition of such a grammatical contrast? Views on this issue vary. Brown (1973) used the 90% correct usage in his study of Adam, Eve, and Sarah's linguistic production. He assumed that the child had acquired a linguistic contrast when she or he produced the functor morpheme in 90% of the cases where an adult English speaker would produce it. Another cutoff point for assuming successful acquisition of functional morphology is 60%,[4] presumably because it is significantly different from chance. Finally, another cutoff point proposal comes from the child language literature, one as low as first productive use in children, followed soon after by regular use (Stromswold 1996, Snyder 2007). The logic of the latter is that if a child produces a certain construction freely and without imitation, and uses it with a variety of lexical items, then this construction has entered the child's grammar. Snyder (2007) gives an example from the acquisition of the verb–particle construction (*She drank the coke up*) by Sarah. At first, two isolated uses of the construction are found in the transcripts but they are not followed by regular use, hence they are not deemed to be indicative of a new construction in the grammar. Soon after

[4] This cutoff was used by Vainikka and Young-Scholten (1994). Note, however, that it is far below adult native speaker norms.

that, a veritable "geyser" of examples of verb plus particle is attested in the transcripts. The age of 2;06,20 (two years, six months, and 20 days) is established as the first of repeated uses (or first clear use, followed soon after by repeated use), and that is considered to be the age when Sarah acquired the verb–particle construction.

While the first of repeated uses seems justified for child language acquisition, the case for using this criterion to gauge successful acquisition in the second language is far from clear-cut. One important difference is that children very rarely make errors with functional morphology: the large majority of their errors are simply omissions. While omissions are also attested in L2 acquisition, there are also the so-called errors of commission, such as using one form of a functor where another one is required. Furthermore, in order to establish whether a functional category has been acquired, we have to take into account not only production of the correct form but also comprehension of the correct meaning. In the next sections, I consider several theoretical proposals on the form–meaning mapping, with respect to functional morphology, in adult L2 acquisition.

7.2 Syntax-before-morphology, White (2003)

In her textbook on generative second language acquisition, Lydia White (2003) frames this debate in an even more interesting light. In addressing the issue of when we can consider that a functional category has been acquired, she argues that another type of information is also crucial to consider: not only surface morphology and grammatical meaning, but also syntactic information related to that category. To take an example from the functional category of Tense Phrase (TP), consider the following example:

(2) He often take **-s** the bus.

Agree

[3rd person, singular subject]
[Tense: present]
[Aspect: habitual]

but also

Overt Subject obligatory
Nominative Subject
Verb stays in Verbal Phrase

Three reflexes of the Functional Category Tense Phrase (TP)
Morphophonology: -*s*;
Meaning: present, habitual event;
Syntactic information: subject agreement, obligatory subject, nominative subject, verb in VP.

The subject pronoun carries the features [masculine], [3rd person], and [singular]. These features are interpretable, that is, they contribute to the meaning of the sentence. The features on the verb ensure agreement with the subject, and so they are not important for understanding the sentence meaning. We call them "uninterpretable".[5] The latter features are extremely important for triggering and establishing syntactic dependencies, (essentially, the case when the form of one category depends on another). According to this fundamental distinction, the presence of an uninterpretable feature in a derivation forces a search for a matching (interpretable) sister feature. When this match is achieved through the process of Agree, the uninterpretable feature is deleted from the derivation. However, it has already executed its function of creating a syntactic dependency, including movement of phrases. Thus, only interpretable features survive into the syntax–semantics interface. In oversimplified terms, interpretable features remain to be taken into account in the meaning calculation while uninterpretable features regulate core syntactic behavior.

However, that's not the only function of the agreement morphology. The morpheme -*s* has additional features: it also signals that the verb is in the present tense and denotes a habitual activity. In addition to agreement, tense, and aspect, a host of other obligatory facts are captured by the features of the tense morphology: the fact that in English the subject is obligatory, that it is in Nominative case, and that the verb stays in VP, as signaled by its being on the left of the VP-edge adverb *often*. All these properties are English-specific; tense inflections in other languages signal different values. For example, the subject in Russian can be Dative, the subject in Spanish and Mandarin can be null (unexpressed), and the main verb in French and Italian go on the other side of adverbs such as *often*. As

[5] Chomsky (1995: 277ff). For an accessible review of this topic, see Adger and Svenonius (2011). See also Adger (2003), Chapter 2, for more examples of features across languages.

the reader can easily establish, the little morpheme -*s* on the verb packs a lot of information which impacts on the form and meaning of the whole sentence. All of that *morphological, semantic, and syntactic* information has to be acquired, in order for the morpheme to be properly acquired. However, it is also conceivable that a learner does not acquire all of that information at the same time.

More generally speaking, the set of functional categories constitutes a submodule of the computational system, namely, the Functional Lexicon (and a very important submodule, at that). Each functional category is associated with a lexical item, or items, specified for the relevant formal features. (For example, TP is associated with -*ed*, -s, and various auxiliaries.) A language's parametric choices come from a blueprint made up of a finite set of features, feature values, and properties. One such property regulates whether a certain feature will induce phrasal movement or not, what we call "strength of features." Learning L2 functional categories comprises acquiring the properties of a set of functional lexicon entries. Part of this knowledge involves syntactic reflexes *superficially unrelated* to the morphophonology of these lexical entries, like displacement of a phrase away from its position of merging. It is precisely this syntactic information that White (2003) and Lardiere (1998a,b) argue is important as an indication that a functional category has been acquired. The logic goes like this: if all of these formal features, including the meaning, construct the functional category, then knowledge of at least one type of information (morphological, syntactic, or semantic) can attest to the successful acquisition of a category.

But which one comes first, knowledge of syntax or accurate suppliance of morphology in production, in the building of the second language grammar? One approach, dubbed "morphology-before-syntax" by White (2003: 182), proposes that acquisition of the actual morphophonological reflexes of the functional category drive acquisition of the syntax (Clahsen, Penke, and Parodi 1973/74, Radford 1990). Proponents of this approach for SLA include Vainikka and Young-Scholten (1996, 1994) and Hawkins (2001). The opposite claim, named "syntax-before-morphology," finds support from data like the following in Table 7.1 (see also Table 6.2 in Chapter 6, White 2003, p. 189).

Evidence for the separation of syntactic knowledge and morphological accuracy comes from child and adult production of L2 English (Lardiere 1998a,b; Li 2012). Lardiere's subject, Patty, is a Hokkien and Mandarin-bilingual adult learner of English (see more in Exercise 7.3). Li's participants

Table 7.1 L2 English suppliance of functional morphology in obligatory contexts (in %)

	3 sg. agreement	Past tense on lexical verbs	Suppletive forms of *be*	Overt subjects	Nom. case	V in VP
Lardiere (1998a,b)	4.5	34.5	90	98	100	100
Li (2012)	16	25.5	93	100	100	–

are six Mandarin-native children aged 7 to 9 acquiring English in a naturalistic environment in the US. Patty's performance is considered to be at end-state, in the sense that she will not develop it further. The children's performance is captured longitudinally for eight months, starting when they had been in the US for four months.

What is especially striking in the data presented in Table 7.1 is the clear dissociation between the incidence of verbal inflection (ranging between 34.5% and 4.5%) and the various syntactic phenomena related to it, like providing overt subjects, marking nominative case on the subject, and the verb staying in VP (above 98% accuracy). It seems that Patty and the children do not produce the overt morphemes -*s* and -*ed*, but they know what the morphemes stand for and what other syntactic processes they regulate in the sentence. Knowledge of all the properties reflected in Table 7.1 is purportedly knowledge related to the same underlying functional category of Tense Phrase and its features. In view of such data, it is hard to maintain that omission of functional morphology is indicative of lack of L2 morphosyntactic features.

Data from acquisition of grammatical meaning also supports the morphology-before-syntax view. In a little twist of the logic of the argument presented above, Slabakova (2003) investigated English L2 knowledge of various meanings associated with the functional categories Tense and Aspect. These meanings included the ongoing interpretation of the (present) progressive tense (*I am eating an apple*), the habitual interpretation of the simple tense (*I eat an apple every day*), and the temporary state meaning of the progressive tense with stative verbs (*I am being lazy today*). The native Bulgarian of the learners does not mark these meanings with verbal inflectional morphology. However, all these meanings are taught in language classrooms, at least in Bulgaria, from the first days of classroom exposure. It could have been the case that English L2 speakers were so accurate on these meanings because they had learned explicit rules about temporal and aspectual interpretation. That is why another, related, meaning was included and tested in the experimental study: the complete event interpretation of bare verbs as in *I saw Jane cross the street.* Jane must have crossed the street to the other sidewalk, for this sentence to be True. This meaning is not taught, but learners were as accurate on it as they were on the other, instructed, meanings. The researcher concluded that once the functional categories are activated, all the attendant meanings come into the grammar of the learners.

7.3 Representational Deficit Hypotheses

The opposite approach to the one described in the section above is to postulate that there is no meaningful separation between the morphological, syntactic, and semantic information making up knowledge of a functional category. The immediate and logical consequence is to assert that inferior or poor performance on functional morphology is indicative of incomplete or non-stable grammatical competence (Johnson, Shenkman, Newport, and Medin 1996). Have we seen this assertion made before? This type of logic is also in evidence in the morphology-before-syntax view and in the morpheme order studies, as discussed above. Recall that we asked the question whether targetlike performance on functional morphology is tantamount to, or at least indicative of, linguistic knowledge. The answer is a qualified yes on the representational deficit view and a qualified no, on the full functional representation view, exemplified by the Missing Surface Inflection view (see next section).

For theories adopting a representational deficit view, learners' problems with inflectional morphology are caused by missing or defective syntactic features that trigger morphological agreement. But what exactly does the representational deficit view hold? Several researchers over the years, notably Roger Hawkins and Ianthi Tsimpli, have argued that abstract morphosyntactic features, including those relevant for agreement and case marking, are only accessible to adult L2 learners if they are instantiated in their native language, but not otherwise (e.g., Hawkins and Casillas 2008, Tsimpli and Dimitrakopoulou 2007). In an effort to discuss morphology and syntax separately, as far as possible, I shall expand on this approach in the next chapter. In this chapter, the discussion is confined to this view's assumption that a strong link exists between inflectional morphology and syntax in L2 grammars. A common thread among representational deficit positions[6] is that morphosyntactic representations of adult L2 learners are somehow impaired and that errors with morphology are indicative of deeper problems with syntax. For example, early approaches to missing inflection in the verbal domain argued that absent or incorrect agreement morphology on verbs in the initial state of SLA reflected either missing functional projections such as Inflectional Phrase (IP, equivalent to TP in later analyses) and Complementizer Phrase (CP) (Vainikka and Young-Scholten 1996, 2013) or unvalued "inert" features in the IP projection (Eubank 1996).

One study that upholds this position with data from inflectional morphology is Hawkins and Liszka (2003). This is a study that compares the incidence of past tense morphology in oral production (a film retell task) by five German, five Japanese, and two Chinese speakers learning English. The Chinese learners produced significantly fewer *-ed* morphemes (63%) as compared to the Japanese and German speakers (92%). According to Hawkins and Liszka, this discrepancy is due to the Chinese learners' impaired representation of the Tense functional category. The reason for this impairment is that, according to some syntactic proposals, the Chinese language lacks TP. However, Hawkins and Liszka did not report on their Chinese learners' knowledge of the other syntactic properties (overt subject, Nominative subject, etc.) and how that knowledge compares with that of the German and Japanese learners'. At issue remains what compels the Chinese learners to produce the past tense morphology in the 63% that they do it correctly. The representational deficit response to such a question is that successful performance is due to superficial noticing of the morpheme and imitation of native speakers. In other words the Chinese learners are "faking" correct performance without a correct underlying representation.

Several further studies by Hawkins and Franceschina (Franceschina 2001, 2005, Hawkins and Franceschina 2004) on the L2 acquisition of gender in the Noun Phrase (NP) are also relevant to this issue. In order to understand the concrete claims of this research, we need to resort to the distinction between

[6] Over the years, the impairment view/approach/position has included the Valueless Features Hypothesis (Eubank 1993/4), the Local Impairment Hypothesis (Beck 1998), the Failed Functional Features Hypothesis (Hawkins and Chan 1997) and the Interpretability Hypothesis (Hawkins and Hattori 2006, Tsimpli, and Dimitrakopoulou 2006).

interpretable and uninterpretable features as defined above. In the area of gender marking, the inherent gender feature of nouns marked in the lexicon is considered to be interpretable. However, the same feature on adjectives agreeing with the noun in the so-called gender concord is uninterpretable. Note that both features are reflected in inflectional morphology. The following examples come from Harris (1991):

(3) a. Mi sobrino/padre (*m*) es inteligente (*m*)/alto (*m*)
'My nephew/father is intelligent/tall.'
b. Mi sobrina/madre (*f*) es inteligente (*f*)/alta (*f*)
'My niece/mother is intelligent/tall.'

The Interpretability Hypothesis (Hawkins and Hattori 2006, Tsimpli and Dimitrakopoulou 2006) makes use of this distinction to argue that interpretable features are acquirable, because they are vital for interpretation, while uninterpretable features are not acquirable, except when they come in the L2 grammar through the native grammar. In their work on nominal features, Hawkins and Franceschina show that even highly advanced L2ers whose L1s do not have grammatical gender experience difficulties acquiring gender concord in L2 Spanish. For example, Franceschina (2001, 2005) reports data from a case study on Martin, an English learner of L2 Spanish, who had supposedly reached a steady state grammar. Martin was nearly natively accurate with gender assignment on nouns (where these features are interpretable); however, he demonstrated considerable variability with gender assignment on determiners and adjectives, favoring the masculine gender as default. Franceschina also reports that Martin was accurate to a native level with number assignment across nouns, determiners, and adjectives. This dissociation of accuracy on number and gender is significant. Since English has grammatical number, but not grammatical gender, she interprets this difference in morphological performance as evidence of the fact that new uninterpretable L2 features cannot be acquired. Thus, both Hawkins and Franceschina argue that the mental representation of grammatical gender for L1 English adult learners of Spanish is necessarily different from native Spanish speakers.[7]

[7] However, see Sagarra and Herschensohn (2013) for findings of the opposite type, namely, intermediate learners of Spanish demonstrating sensitivity to gender concord, as attested by a processing task.

7.4 The Missing Surface Inflection Hypothesis

The opposing view of L2 speakers being impaired in their linguistic representations can be dubbed Full Functional Representation. There are several proposals within this position that we will examine in the next sections. The common thread between these hypotheses is that it is deemed possible for second language learners to acquire nativelike functional representations and thus achieve grammars that are not *qualitatively* different from those of native speakers, though they may be *quantitatively* different. For example, in Section 7.2, we were able to appreciate the fact that inconsistent, seemingly random suppliance of inflectional morphology might underrepresent actual knowledge of functional categories. I argued, following White and Lardiere, that knowledge of syntactic and semantic reflexes of a functional category also attests to knowledge of the category itself, because all three types of features (morphological, syntactic, and semantic) go together to construct the category. This is essentially a charitable view of second language knowledge: speakers actually know more than we give them credit for.

There is no doubt that *morphosyntactic variation* is the biggest challenge to the view that L2 acquisition is possible and attainable in adulthood. Individuals vary on the correct suppliance of (the morphophonology of) inflectional affixes, even if they are at the same level of proficiency. They vary in the accurate production of the morphology depending on the time of the day and whether they have had coffee that morning. Patty, Donna Lardiere's subject, is a highly functional English speaker and a professional with a successful career in a US company, but she produces the past tense ending only 35% of the time. In comparison, children rarely make mistakes with such affixes, and after they acquire a certain functional category, it seems that they know all there is to know about it. (Sometimes it just seems so, but that's another story.)

Pervasive and extensive variation is what all second language researchers have to explain, then. We should also keep in mind that variation is most visible in production, but that may not be the whole story. The Missing Surface Inflection Hypothesis (MSIH) (Haznedar and Schwartz 1997, Prevost and White 2000) provides one explanation of the relationship between suppliance of functional morphology and knowledge of the syntactic properties underlying this morphology, or associated with this morphology's feature bundle. This theory is, in a way, a direct extension of the syntax-before-morphology view discussed above. Under this view, there may be no representational deficits in learners whose language production of the morphology is not optimal. However, there may be a mapping problem between abstract features and surface morphological forms, such that incorrect production underrepresents underlying knowledge. In a nutshell, some rupture occurs between syntax and morphology such that the morphology is somehow missing, but only on the surface. Hence the name of this hypothesis.

Evidence for this view comes again from child and adult learners. Haznedar and Schwartz (1997) reported findings from the production of a four-year-old Turkish-speaking child, Erdem. Erdem's development in English was followed with regular recordings over 18 months. In child first language acquisition, producing finite verbal forms is related to the obligatory suppliance of overt subjects, in the sense that finite verbal forms and overt subjects have to come together in learner grammar. Erdem started producing subjects long before his use of verbal inflection in obligatory contexts rose to nativelike levels. This dissociation suggests that the relationship between the use of inflectional morphology and overt subjects is broken, at least for him. Haznedar (2001) presents robust evidence for his use of nominative pronoun subjects, suggesting that Erdem's case assignment works well. Haznedar and Schwartz argue that the uninflected forms produced by Erdem are in fact finite forms and they function as such, but that their overt inflections are missing.

Prévost and White (2000a,b) examine similar evidence from the production of children and adults acquiring French and German in naturalistic ways (two Moroccan-Arabic learners of French and two native speakers of Spanish and Portuguese learning German). Again, the distribution of finite and nonfinite verbal forms is under investigation, in its relation with overt subjects (clitics and strong pronouns), as well as negation. Very simply expressed, if learners have knowledge of finiteness, finite forms should not occur in nonfinite positions. The researchers found that their adult learners did not typically use finite forms in nonfinite positions. It is also true that the learners used infinitive verb forms in both nonfinite and finite positions, suggesting that infinitive verb forms are unspecified for finiteness. Thus, the argument is that adult L2 learners have an abstract representation for finiteness that is concealed by their "misuse" of French/German infinitive forms. These results converge with Haznedar and Schwartz's findings.

The question arises of what grammatical process allows this sort of separation between unimpaired syntactic knowledge and haphazard morphological realization. Note that in the case of English-acquiring Erdem, the nonfinite verb forms were just bare verbs, so inflection was truly missing. In the case of Prévost and White's German and French acquiring subjects, however, the nonfinite verbal forms do carry infinitival morphology. Prévost and White explain the separation, or dissociation, they uncover as a mapping problem between the lexical items of the inflectional affixes and the feature specifications of the syntactic nodes where the inflected forms are inserted.

In order to understand this claim, we need to take an excursion into a linguistic proposal called Distributed Morphology (Halle and Marantz 1993) and more specifically, its claim of late lexical insertion. According to this proposal, the syntactic tree for the sentence to be produced provides terminal nodes in which all morphosyntactic features have been checked and validated (different approaches to feature checking would have different validation mechanisms). For example, if the subject is *John*, the verbal form has to be specified for 3rd person singular features, among others (*John like-**s** sushi*). The next step in the derivation involves competition between fully specified lexical items for insertion into the syntactic nodes. An inflected (finite) form such as *like-s* is associated with grammatical features such as tense, person, and number (in some languages even gender) as part of its lexical specification. In lexical insertion, the features of a lexical item should be consistent with the features of the syntactic node where it is inserted, but the most highly specified item wins the competition for insertion. The requirement is that the features of the syntax node and the lexical item match, but the latter can also be a subset of the former. In the case of mapping breakdown, learners may not have easy access to the fully specified best candidate for lexical insertion, and they may instead access a less specified but more accessible item.

For Prévost and White (2000), L2 learners acquire the relevant grammatical features of the terminal node in the syntax via the native language, UG or the L2 input, but they might not have fully acquired the feature specifications of the associated items from the functional lexicon. Verbal paradigms are hard to learn, and in some languages more than others. Prévost and White propose that learners sometimes reach for "default forms" such as infinitives because they are easy to access in their mental lexicons, being basic and frequent. Under this analysis then, there is no syntactic deficit in interlanguage grammars but performance limitations, sometimes resulting from communication pressures, may lead to omission of inflection or use of default forms (2000a: 129). In other words, failure to produce consistent inflection is attributed to difficulties in accessing the relevant lexical items by which inflection is realized, particularly when speaking.

Note that an immediate prediction of this account is the following: omission and use of default forms should happen more in production than in comprehension, and more when the learner is tired, distracted, or under some communicative pressure. This is because variability in morphology is linked to lexical access, which is arguably more difficult in production. McCarthy (2008) set up an experimental study to check the former prediction. She tested English learners of Spanish on production and comprehension of Spanish object clitics, and looked at agreement in gender and number. McCarthy discovered that for her intermediate learners, morphological variability was attested in production but extended to comprehension as well. She argued that there may be representational problems with the inflectional morphology, as well as access problems, for these learners. Furthermore, she documented the use of default forms in production as well as comprehension. For instance, the masculine object clitic *lo* was more likely to be interpreted as referring to a feminine object. However, McCarthy did not uncover such variability in the comprehension of the feature number, and the use of the singular as default was only attested in production. Furthermore, the acquisition of the gender feature has proved very difficult for learners (Unsworth 2008a), probably because there is a lot of lexical learning involved in acquiring the gender specification of each noun one by one.

Support for McCarthy's findings comes from a recent processing study, Reichle, Tremblay, and Coughlin (2013). The researchers measured ERP responses of intermediate learners of French reading sentences in which subject–verb agreement was violated. In an interesting twist, the researchers employed two conditions: in one the agreement violation was local (the subject and verb were close together); in the other it was long distance (the subject and verb were separated by another nominal). They also tested the working memory span of the learners. The learners' responses revealed N400 effects when they encountered subject–verb agreement violations. This ERP effect is a well-known response to lexical and semantic violations, suggesting that the learners did not treat these as morphosyntactic violations (as they did in their native English). Furthermore, the size of the L2 learners' N400 effect decreased when the subject noun was not adjacent to the verb as compared to when it was. The latter type of sentence taxes working memory and increases its load. The authors found that:

> L2 learners who excel at performing simultaneous linguistic operations such as reading sentences for meaning and holding sentence-final words in memory (i.e., reading-span task) also excel at holding subject nouns in memory when processing non-adjacent agreement dependencies (i.e., ERP task).

These findings indicate that, at intermediate proficiency levels at least, mental lexicon access may be involved in processing agreement morphology, modulated by learners' working memory capacity.

In summary, the MSIH capitalizes on the three types of linguistic knowledge that together comprise knowledge of a functional category: the morphophonological, syntactic, and semantic reflexes of that category. The hypothesis proposes that even if the syntactic side of a functional category has been acquired, the morphological forms may not be always supplied adequately in production, due to a mapping issue between the syntax and the functional mental lexicon. A substantial body of research from child and adult learners lends support to this separation of syntactic and morphological knowledge (e.g., Hyams 1994). McCarthy's (2008) research suggests that problems with the morphology may affect comprehension as well, but only among intermediate learners. Research that aims at directly supporting the MSIH should demonstrate higher variability for inflectional morphology in the same individuals, under different production and comprehension conditions.

Teaching relevance

It might be helpful for teachers to know that failure to produce correct inflectional endings is a work in progress. Such failure may not reflect underlying problems with grammatical information. Errors may be more pronounced in production than in comprehension. Since the problems are of a superficial nature and are related to lexical retrieval, ample practice of grammatical morphemes in language classrooms is warranted. Such practice will improve the automatic lexical access for function words and inflections.

7.5 The Prosodic Transfer Hypothesis

Still within the Full Functional Representation position, another proposal to at least partially explain morphosyntactic variation is Goad and White's Prosodic Transfer Hypothesis (PTH) (Goad, White, and Steele 2003, Goad and White 2004, 2006). It is compatible with the MSIH in the sense that it offers another reason why learners might not produce verb endings or other functional morphology. Just as the MSIH, the PTH suggests that L2 learners' syntactic representations are appropriate for the target language (that is, there is no representational deficit). However, L2 learners may delete functional material in production, since the prosodic structure of their L1 may differ from the L2, leaving no way of building the correct prosodic structure for the L2.

But let's look at prosodic structure and how it may be relevant for second language acquisition (Nespor and Vogel 1986, Selkirk 1984, 1996). It should come as no surprise to the reader of this book that phonological structure is proposed to be hierarchical, in the sense that units merge together to form bigger and qualitatively different units. Scholars of phonology have discovered that segments are organized into a syllable, syllables are organized into a foot, feet are organized into a prosodic word (PWd), and prosodic words are organized into a phonological phrase (PPh). Note the resemblance to syntactic structure trees.

(4) Phonological Phrase (PPh)
|
Prosodic Word (PWd)
|
Foot (Ft)
|
Syllable (σ)

Where a certain sound or a certain affix is attached within the prosodic word or the phonological phrase is important for language acquisition. For example, children pronounce coda consonants more often (instead of dropping them) in syllables of longer duration. Since grammatical affixes, especially in English, frequently consist of a single sound /s, z, t, d/, prosodic effects are attested on the acquisition of the functional morphology. A recent experiment (Sundara, Demuth, and Kuhl 2011) discovered that 2-year-olds noticed the difference between grammatical (e.g., *Now she cries*) and ungrammatical sentences missing the agreement marker *-s* (e.g., *Now she cry*). However, when the agreement marker was inside the sentence (*She cries now* vs. *She cry now*), the same children did not show sensitivity to the difference. Clearly, the position of the functional morphology determines its perception. In another example, 2;3-year-olds were shown to be more likely to produce the definite article when it followed a monosyllabic word (e.g., *Tommy kicks the rabbit*) than a disyllabic verb (*Tommy catches the rabbit*). The explanations of both findings have to do with the prosodic structure of the utterance and its effect on acquisition, perception, and production of grammatical morphology. Scholars of child language (e.g., Demuth 2001, 2014) argue that the variable emergence of early grammatical morphemes are *prosodically licensed,* as well as grammatically licensed.

It is hardly surprising, then, that such effects are being discovered in second language acquisition as well. This is exactly what Goad and White (2004, 2006, 2008) propose. Languages differ in the way they prosodify (attribute prosodic structure to) functional morphemes, and differences between the L1 and L2 may turn out to be significant for production of affixes. To date, these researchers have studied production and comprehension of functional morphology (past tense, agreement, articles) in Chinese-English and Turkish-English interlanguage. The main idea is as follows: learners may not produce inflectional morphology while they have fully engaged the functional category of that morphology and obey all the syntactic and interpretive consequences of that successful acquisition.

To take an example from past tense morphology, Goad and White (2006) start by showing that the prosodic structure of words ending in consonant clusters is different for regular and irregular past tense expression.

(5) a. English regular past tense   b. English irregular past tense

In English, the regular *-ed* ending as in (5a) is realized as adjoined to the prosodic word, while the irregular ending, that looks similar, actually has a different structure, all organized within the prosodic word. In Mandarin, on the other hand, the only inflection is aspectual, and it is organized within the prosodic word, just as in English irregular past tense.

(6) PWd

mai3 lə5
buy PERF 'bought already'

Assuming that the prosodic structure required to pronounce English past tense endings and past participles is absent in Mandarin, Goad and White predicted that initial learners' production accuracy would be depressed. However, at later stages of development, learners will be able to minimally adapt their native prosodic structure to new configurations and will be able to pronounce the endings. Compare the predictions of the representational deficit account with respect to the production of past tense endings. If learners have impaired past tense representations, it will make no difference whether they are trying to pronounce regular or irregular verbs: they will be equally incorrect. Hawkins and Liszka (2003) acknowledge that this prediction is not supported by their data.

Goad and White (2006) administered an interpretation test and a production task on the same stimuli. First, participants had to choose the correct continuation (past or present) for a past adverbial lead in, as follows:

(7) Last night after dinner,
you showed me photos of your daughter.
you show me photos of your daughter.

The ten learners, who were of intermediate proficiency, achieved 83% accuracy in choosing the correct continuation. After they had registered their choice, they were asked to read it out loud. Only correct choices were counted in the second task. Results show that learners were highly accurate on the production task as well, with accuracy percentages between 87 and 94 on the various types of stems. No significant differences emerged between regular and irregular past tense forms. Although the prediction of the PTH for a differential accuracy on regulars and irregulars was not supported by this group of subjects, it is entirely possible that learners at earlier levels of development could show such support. Importantly, these Mandarin learners of English demonstrated both high levels of suppliance of morphology as well as superior knowledge of interpretation, which are two clues for successful acquisition. Goad and White (2008) offered a detailed examination of the phonetics of learners' past tense and article production. They showed that many of the learners' outputs cannot be due to what the second language input offers them. In particular, learners produce tense and articles in ways that are not found in either the L1 or the L2 (fortis release on tense inflection and stress on articles). The researchers argue that transfer from native prosodic structures, which learners then try to adapt to the target prosodic structures, explains their findings best.

In sum, some (although not all) of the variation in functional morphology production may be due to the necessity to adapt native prosodic structures to target language prosodic structures. UG supplies the template of how to do this, but it is a learning process that takes time. Goad and White specifically argue that their hypothesis does not apply to perception; in other words, prosody does not act as a filter for comprehension.

7.6 The Feature Reassembly Hypothesis

Lardiere's Feature Reassembly Hypothesis (FRH) (Lardiere 2009a,b) is a theoretical proposal that builds on the observation that successful acquisition involves acquiring the set of formal features of the second language. As we remember, formal features include phonological, syntactic, and semantic features, bundled together on the lexical items of every language.[8] Languages differ in what features they encode in the various pieces of functional morphology. Where do all these features live? A mainstay of the Minimalist Program is the proposal that features come from a universal repository, made available to the child together with the computational procedures of how to combine these features on lexical items, check and delete uninterpretable features, etc.

Of course, acquiring a second language involves acquiring a second set of features, different from the native one. It is often the case, as Lardiere (2007,

[8] Borer (1984), Chomsky (1995): the so-called Borer–Chomsky Conjecture (Baker 2008).

2009a) notices, that "[a]ssembling the particular lexical items of a second language requires that the learner reconfigure features from the way these are represented in the first language into new formal configurations on possibly quite different types of lexical items in the L2." (Lardiere 2009a: 173). Thus, learning lexical items with bundles of features in possibly new configurations appears to be the most important learning task. I shall illustrate this with one extended example from the feature [±past] (Lardiere 2007, Slabakova 2008). This feature captures the temporal relationship between the event and the speech moment: past events happened before the speech moment; non-past events either hold at the moment of speech or in a future time interval. Languages can choose to represent or not represent this feature with an overt morpheme. The oft-cited languages that differ in this respect are Mandarin Chinese, without past tense marking, and English, with past tense marking. However, the mapping between meaning and form in English is considerably more varied than this simple, one-to-one mapping configuration may suggest. As Lardiere (2005) points out, it is a many-to-many mapping. First of all, the *-ed* morphology encodes perfective aspect in English (Smith 1997), as the example in (8) demonstrates.

(8) We devoured the pizza on the terrace (and there is no more pizza left).

The pizza-devouring event in this sentence is not only in the past, it is also complete, so uttering the second clause in (8) as a continuation of the first is not a contradiction.[9] Secondly, the past morphology is used as a marker of uncertainty and hence politeness in requests:

(9) I was wondering whether you would be free tomorrow.

In (9), both past tense forms *was –ing* and *would* stand for a present state ("I am wondering as I speak") and a future state ("are you going to be free tomorrow?").

Thirdly, past morphology can be used in sequence of tenses to refer to a non-past state or event, leading to ambiguity.

[9] To understand this, compare the sentence in (8) with its progressive alternative:
(8′) We were devouring the pizza on the terrace when he came.
In (8′), there is no guarantee that the pizza is all gone.

(10) Jane said that Joyce was pregnant.

The utterance in (10) can mean either that Joyce was pregnant at the time when Jane and Joyce saw each other, but Joyce has given birth since; or it can mean that Joyce is still pregnant today, at present. In this utterance the past form is actually ambiguous between a present and a past temporal reference. Note that all these meanings of the past morphology are not equally frequent in the input, learners may not encounter some of the meanings every day, and as a consequence, might not learn all of the meanings at the same time.

Finally, historical present usages are very common in everyday speech and in fiction. The following excerpt is from Marc Nesbitt's short story "Gigantic."[10]

(11) *"I rake dead bats from the hay floor of the bat cage and throw them in a black plastic bag.... I pick up a sign that says, "Quiet! ______ sleeping!," slip in the "Bats" panel, and place it up front, where all the kids can see it. An hour till we open, I go see our one elephant, Clarice."*

Note that all the events described by the underlined verbs denote a sequence of closed events in the past and not ongoing, future, or incomplete events. This usage of the present simple (a.k.a. the historical present) is a pragmatic convention of English (Smith 1997).

Sticking to English past tense morphology and barely scratching the surface, I have demonstrated, following Lardiere (2005, 2007, 2008), that -*ed* in English can encode [+past] but also [–past], politeness, *irrealis* mood in conditionals, and perfective aspect in events, among other grammatical and pragmatic meanings. On the other hand, the simple present tense can also encode a sequence of complete events in the past. Thus, a mapping of many-to-many exists between English past tense meanings and forms.

A further comparison between English, Irish, and Somali points to another source of mismatch: the lexical categories on which a feature is expressed may differ across languages.[11] Irish expresses the tense distinction on its complementizers in agreement with the tense in the embedded

[10] Published in *The New Yorker* on July 9, 2001, p. 76.

[11] The examples are from McCloskey (1979) and Lecarme (2003, 2004), as cited in Lardiere (2005: 180).

clause as in (12) while in Somali [past] is encoded on nouns within the noun phrase, (13).

(12)	Deir	sé	gurL	thuig	sé	an scéal	Irish
	says	he	that-Past	understood	he	the story	

‘He says that he understood the story.’

(13) a.	árday-gii	hore	Somali
	student-Det.Masc.Past	before	

‘the former student’

b.	(weligay)	dúhur-kii	baan	wax	cunaa
	(always)	noon-Det.Masc.Past	Fem.1S	thing	eat-Pres

‘I always eat at noon.’

c.	Inán-tii	hálkée	bay	joogta?
	girl-Det. Fem.Past	place-Det. Masc.Q	Fem.3S	stay-Fem. Pres

‘Where is the girl?’

The mismatches don’t stop here. The expressions of the *-ed* past morpheme in English are phonologically determined: they surface as /-t/, /-d/ or /-ɪd/ depending on the preceding sound. Furthermore, if one considers the frequent irregular verbs with their (partly) phonologically determined classes of past tense marking, it becomes abundantly clear that the one form–one meaning mapping in the functional morphology of human languages is very rare indeed. Thus what are crucial to acquire are not only the meanings and the morphophonological forms of the affixes but also the conditioning environments in which they appear.[12]

Research applying the predictions of the FRH to the acquisition of second language properties is just beginning to pick up steam. However, this hypothesis has engaged the imagination of SLA scholars[13] and research is proceeding apace. In fact, this is one of the more promising future directions for SLA research, because it compels researchers to build a

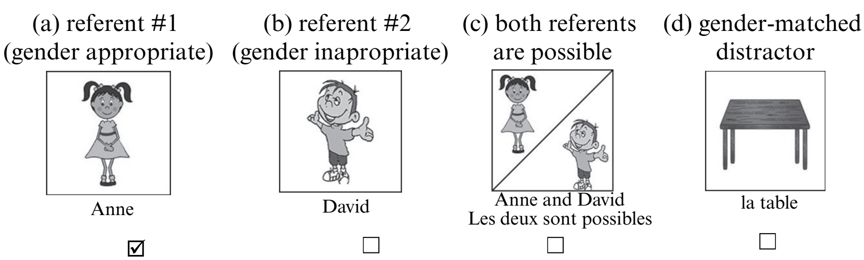
[12] Lardiere (2009a) provides an extended comparison between Mandarin and English plural, and the reader is encouraged to tackle the original article at this point.

[13] See Dominguez, Arche, and Myles (2011), Gil and Marsden (2013), Cho and Slabakova (2014), Guijarro-Fuentes (2011, 2012), Shimanskaya and Slabakova (2014).

more complex picture of the acquisition process and to pay attention to meaning as well as syntactic expression, conditioning environments of the functional morphology, learning features of the same affix one by one, and so on. I shall take one example here from Shimanskaya (2015) (see Shimanskaya and Slabakova 2014), a study which looks at personal pronouns in English-French interlanguage.

English and French personal pronouns differ in many respects, the most visible one being that French has clitic pronouns that attach to the left of the verb (*Je l'aime* 'I love him'). Moreover, while English encodes gender only in the case of [+Human] referents (biological gender), French uniformly encodes grammatical gender in determiners, accusative clitics, and strong pronouns. A distinction not studied before involves another feature: while English pronouns *lexically encode* the [±Human] distinction (*him/her* versus *it*), in French the same accusative clitics (*le, la*) are used to refer to [+Human] and [−Human] antecedents. In the sentence *Je* ***la*** *vois* 'I see her/it' the feminine clitic *la* can refer to *la table* 'the table' and *la fille* 'the girl'. Note how in English we have to choose between *I saw her* and *I saw it*. Thus, the feature [±Human] is active in the English pronominal system but is not expressed in the French pronominal system. It was predicted that since the L1 lexically encodes the feature, anglophone L2 learners would initially attempt to establish a similar contrast in the L2, and they will transfer this feature from the L1. Three groups of French learners at different levels of proficiency took a forced-choice picture selection task (among others), in which they read context sentences as in (14), after which they had to choose which picture corresponded to the test sentence.

(14) Context: On Tuesday evening, Nicolas goes to the library to meet Anne and David.
Nicolas: Parfois, je **la** vois près de la fenêtre. (Sometimes I see **her** near the window)



The test sentence contains the feminine clitic *la* and, thus, can only be interpreted as referring to *Anne* (or the distractor *the table*). A gender error in the example above would be choosing David.

Animate and inanimate referents were counterbalanced in the experimental design. Beginning and intermediate learners made significantly more gender errors with inanimate than with animate nouns. This pattern of behavior suggests that the learners experience relatively more difficulty in using grammatical gender to refer to inanimate nouns. The natural gender in their native English (*he/she*) helps them with the grammatical gender in French, but only within the set of humans at first. Biological gender takes precedence over grammatical gender when disambiguating pronominal reference in L2 learners whose L1 does not have grammatical gender. By advanced proficiency levels, the new feature of grammatical gender is added to the L2 grammar, and accuracy reaches 99%. The take-home message of these experimental results is that the reassembly process is not instantaneous, and that it is undoubtedly influenced by the native features.

In conclusion, we have seen in this chapter that learning the functional morphology in a second language is crucial because it is where languages differ. This acquisition is complicated because a functional meaning may be represented on one lexical category in the native language and on another lexical category in the second language, or not at all overtly represented (cf. Irish, English, and Somali expressions of Tense). It is also complicated because a functional meaning, say 'past,' can be expressed with dedicated morphology in one language, while in another language it is expressed only optionally or not expressed at all; it can also be expressed with forms lacking this morphology (the historical present). Finally, acquiring a functional form is difficult, even if it has constant meaning, because it may have varied expressions depending on different phonological and morphological environments. As the form–function mismatches discussed above attest, the learning task involves much more than a simple pairing between meanings and pieces of functional morphology.

I have argued that the current findings advocate for two (tentative) conclusions. First, even if the functional morphology is not produced with high accuracy, this is not a necessary indication of impaired underlying representations but can be due to lexical access difficulties (the MSIH) or prosodic constraints on pronunciation (the PTH). Secondly, it is possible that getting together the complete feature bundle of an L2 functional category is a nonlinear (hence fragmented) and extended process, affected by the native feature composition and the evidence for each feature in the input. Since acquisition of the functional morphology is most important for correct production and comprehension in the L2, it should get the lion's share of attention in language classrooms.

7.7 Exercises

Exercise 7.1. Here are some explanations of the morpheme order findings:

a) This is a natural acquisition order of acquiring English grammatical morphemes. Since it is guided by universal principles, it does not depend much on the native language (e.g., Krashen 1977).
b) It is a combination of five factors (perceptual salience, semantic complexity, morphophonological regularity, syntactic category, and frequency) that predicts about 71% of the variance observed in the studies included in Goldschneider and DeKeyser's (2001) meta-analysis.
c) The influence of the native grammar has been underestimated so far. It does exist (Luk and Shirai, 2009).
d) Learners are building a mental grammar of the L2 (in this case English), and the order reflects the hierarchical organization of the clause structure that is being acquired: CP-over-TP-over-VP (Hawkins 2001).

Discuss these explanations in turn, paying attention to whether they are mutually exclusive or not. Can you find a way to reconcile explanations c) and d)? Can you see the connection between explanations a) and d)?

Exercise 7.2. Debate the relative merits and shortcomings of the different cutoff points for accepting that a piece of functional morphology (a functor, in Brown's terminology) has been acquired. Recall that 90% and 60% of required contexts have been proposed, as well as evidence of first productive (non-chunk) use.

Exercise 7.3. In this exercise, we will scrutinize the production of Patty (not her real name), the fossilized learner whose linguistic production is

Table 7.2 Summary of Patty's English

Structure	**Use**	**Grammatical property in Hokkien and Mandarin**	**Has she learned it?**
Present tense -*s* (3sg subject)	4–5%	No overt tense morphology	
Past tense -*ed*	35%	No overt tense morphology	
Case marking	Nativelike	No case marking	
Overt subjects	100%	Null subjects allowed	
Definite determiner	84%	No articles	
Question formation	Nativelike	Wh-in-situ	
Passives	81%	Passives are marked differently	
Possessive pronouns	Near nativelike	With a particle	
Plural marking	84%	No overt morphology	

described in Lardiere (1998a,b; 2007). Patty's characteristics as an L2 learner are as follows (see Table 7.2[14]):

- Hokkien and Mandarin bilingual speaker (dominant in Hokkien), English is (at least) her fourth language.
- Resident in US since her early 20s.
- Company executive, master's degree.
- Married to a Vietnamese L1 speaker and later to an English L1 speaker.
- Recorded at age 32 (after ten years of residence in the US) and at age 40 (after 18 years of residence)

Some examples of Patty's English include:

1. I feel like I should buy the other one too.
2. Something that have to show the unbeliever that you are in spirit.
3. He have the inspiration to say what he want to say.
4. I don't know if everybody can speak in tongue.
5. But I know that I have doubt.
6. That's the gift of wisdom that we have to manifest.

[14] Thanks go to Laura Domínguez for allowing me to use the summary table she created.

7. Everyone who believe it can get it.
8. When you pray what you want you got what you want.
9. Yesterday they open until five.
10. He call me last night.
11. She give me a lot of help.
12. Why do you want me to go?
13. You don't know who you should associate with.
14. She was chosen from forty people.
15. We spoke English to her.
16. He make me uh spending money.
17. She keep asking me to get a concert . . . and asked him to go to this place.
18. And uh that's why M, at that time M want to go too, but her, her sister said no, that's why she never come.
19. They kick him.

Discuss Patty's production, using both the table and the examples. Find examples that would support each line of the table.

Look at the second column of the table. Can you assert that Patty's production is affected ONLY by her native languages?

Discuss what you would answer in the last column of the table, for each property. Reconsider your answer keeping in mind complete acquisition of functional categories (which includes morphological, semantic, and syntactic reflexes).

Exercise 7.4. Based on Montrul (2011). Montrul (2011) is a study comparing the morphological production of adult anglophone L2 learners and heritage speakers of Spanish residing in the US. I will provide some of the findings below, so that you can answer the following questions.

To test production and affix recognition in the grammatical area of tense and aspect, Montrul used an oral and a written task. The oral task involved pictures of Little Red Riding Hood and asked participants to retell the story in the past. The written task was a forced choice of preterit and imperfect morphology (so chance was 50%) in a story told in the past. Here is an example:

(i) El jefe le **daba/dio** el dinero a la empleada para depositarlo en el banco. La empleada **trabajó/trabajaba** para la compañía pero no **estuvo/estaba** contenta con su trabajo y **quiso/quería** otro trabajo. La mujer **necesitó/necesitaba** salir del pueblo.

Table 7.3 Percentage accuracy on tense–aspect and mood morphology in the written recognition and oral production tasks

Group	Tense–aspect		Mood	
	Preterit	**Imperfect**	**Indicative**	**Subjunctive**
L2 learners (n = 72)				
Written recognition	87.1	84.2	87.8	72.5
Oral production	94.3	88.0	86.2	42.6
Heritage speakers (n = 70)				
Written recognition	85.7	70.1	83.1	65.7
Oral production	98.2	95.0	93.3	69.6
Native speakers (n = 22)				
Written recognition	98.9	97.4	99.1	98.3
Oral production	100	100	100	100

'The boss **gave** the money to the employee to have it deposited in the bank. The employee **worked** for the company but **was** not happy with her job and **wanted** another job. The woman **needed** to leave the town.'

To test production and comprehension of indicative and subjunctive mood, the latter unattested in English as widely as in Spanish, she used broad questions designed to elicit opinions and advice (e.g., *What are your plans after graduation and what is your ideal job? What advice would you give to a friend of yours who is an alcoholic? What would you recommend he/she do?*) Some of the contexts required the use of subjunctive forms, while in others it was optional. The written task checking recognition of the mood distinction was in the same format as example (i) above. Table 7.3 presents accuracy percentages.

Questions for you.
Is the oral production of correct functional morphology across the three experimental groups comparable? Is the written recognition comparable? Which group is better at what? Why do you think this is the case?
Discuss the effects of the native language on the L2 learners. Discuss the effect of the dominant English language on the Spanish heritage speakers.
Discuss the possible effect of classroom instruction on the results. Discuss the possible effects of linguistic experience on the differential accuracy.
Are these findings in support of representational deficit hypotheses? Are these findings in support of the Missing Surface Inflection Hypothesis? Are

these findings in support of any other hypothesis we have discussed in this chapter? Support your answers with concrete data.

Can we say that one of the experimental groups has superior morphological competence? Which task is better at capturing this competence?

Find the original study and compare your answers with Montrul's discussion of her results.

Exercise 7.5. Find H. Hopp (2013), Grammatical gender in adult L2 acquisition: Relations between lexical and syntactic variability. *Second Language Research* 29: 33–56, doi:10.1177/0267658312461803.

Download it to your hard drive. The author argues that his "findings support lexical and computational accounts of L2 inflectional variability and argue against models positing representational deficits in morphosyntax in late L2 acquisition and processing." Does he mean the MSIH or some similar account? Make a list of his findings that support this interpretation.